

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester-VI Examination, 2022-23

Botany

Paper: BOB-602 — Microbiology and Plant Pathology

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting two questions from each Section-A and B. Question No. 1 is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following in not more than 50 words each: [1 × 10 = 10]

- (a) What is Hfr cell?
- (b) What is *Halobacterium*?
- (c) What is the difference between lag and log phases of bacterial growth?
- (d) Mention the contribution of Carl Woese.
- (e) What is Mycoplasma?
- (f) What are biotrophic pathogens? Give an example.
- (g) Mention the contribution of Henry Phipps in Indian Plant Pathology.
- (h) What is PTI?
- (i) Name two insect fungi that are used as bio-control agents.
- (j) Name a plant disease caused by phytoplasma.

SECTION A

Question 2. Describe the mechanism of transduction in bacteria with well-labelled diagrams. [15]

Question 3. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Free-living and symbiotic microorganisms
- (b) Cyanophage
- (c) Lactic acid fermentation by bacteria

Question 4. Give a detailed account of the interaction of *Rhizobium* with a leguminous plant. [15]

Question 5. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Compare between Archaea and Bacteria
- (b) Compare between Lytic and Lysogenic cycle
- (c) Plant growth promoting rhizobacteria

SECTION B

Question 6. Briefly describe the various modes of infection in plants caused by fungal pathogens with suitable examples. [15]

Question 7. What are the different types of induced biochemical defense in plants? Describe briefly the role of PR proteins with suitable examples. [15]

Question 8. Answer any two of the following in brief: [7.5 × 2 = 15]

- (a) Causal organism and symptoms of the wilt of tomato
- (b) Modes of the infection of fungal pathogens
- (c) Downy mildew of crucifers

Question 9. Answer any two of the following in brief: [7.5 × 2 = 15]

- (a) Smut of Bajra
- (b) Integrated disease management
- (c) Transmission pathways of the sugarcane mosaic disease

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